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ESSnet Big Data II

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Workpackage WPA

Coordination and Communication

Deliverable A4: report on BDES 2020 and BDES 2021

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Introduction

This is deliverable A4, the report on the BDES 2020 conference and the BDES 2021 roundtable, of the grant agreement of the ESSnet Big Data II. Originally, only one BDES event was planned, BDES 2020, as a physical conference. Due to the pandemic, BDES could not take place in a physical format, and it was decided to split the event in a BDES 2020 conference, aimed at dissemination of the results of the ESSnet Big Data II, and a BDES 2021 roundtable, aimed at reflection on the way forward, given these results. For this purpose, an amendment to the grant agreement was proposed and agreed on. As it turned out, BDES 2021 had to be held in virtual format as well.

The report on BDES 2020 starts at page 3, the report on BDES 2021 on page 16.

Big Data for European Statistics (BDES 2020) – online session

23-30 November 2020

MINUTES

The virtual conference gathered 421 persons from more than 95 institutions from 39 countries. The participants represented international organizations (including UN ESCAP, OECD), European statistical offices (ESS statistical offices), research institutes and financial institutions.

The conference was held on the Zoom platform and was chaired by Marc Debusschere and Peter Struijs.

Monday, 23 November 2020

Session 1 - Introduction

(10:00-12:00 CET)

Peter Struijs (CBS, Statistics Netherlands, NL), the ESSnet Big Data 2018-2020 project coordinator, opened the conference and greeted the participants. He presented main assumptions and the premises for the implementation of big data, including: more accurate statistics, early estimates, faster data delivery to recipients, higher efficiency of statistical research with simultaneous reduction of its costs. Then he proceeded to describe the organizational structure of the project and pointed out countries participating in the project. Next, three tracks under ESSnet Big Data II 2018-2020 were indicated: Implementation (topics: Online Job Vacancies, Enterprise Characteristics, Smart Energy, Tracking Ships and Process and Architecture), New pilot projects (topics: Financial Transaction Data, Earth Observation, Mobile Network Data, Innovative Tourism Statistics and Methodology and Quality), and Preparing smart statistics (Smart Farming, Smart Cities, Smart Devices, Smart Traffic). The participants were also informed that all documents related to the project are published on the project website. At the end of the speech, the project coordinator thanked Martin van Sebille (Project Secretary), Lilli Japec, Giulio Barcaroli, Florian Keusch, Boriska Tóth (members of the Review Board), Albrecht Wirthmann (Eurostat Project Officer), Marc Debusschere, Remco Paulussen and Dominika Nowak (conference co-organizers) for their cooperation.

Then, he gave the floor to the leaders of individual work packages, who briefly presented the topics covered by the project:

- Tomaž Špeh (SURS, Statistics Slovenia, SI) - WPB Online job vacancies
- Galya Stateva (NSI, Statistics Bulgaria, BG) - WPC Enterprise characteristics
- Arko Kesküla (Statistics Estonia, EE) - WPD Smart energy
- Remco Paulussen (CBS, Statistics Netherlands, NL) - WPE Tracking ships
- Monica Scannapieco (ISTAT, Statistics Italy, IT) - WPF Process and architecture
- Johan Fosen (SSB, Statistics Norway, NO) - WPG Financial transaction data
- Marek Morze (GUS, Statistics Poland, PL) - WPH Earth observation
- David Salgado (INE, Statistics Spain, ES) - WPI Mobile network data
- Marek Cierpiał-Wolan (GUS, Statistics Poland, PL) - WPJ Innovative tourism statistics
- Alexander Kowarik (Statistics Austria, AT) - WPK Methodology and quality
- Natalie Rosenski (Destatis, Statistics Germany, DE) - WPL Preparing smart statistics

Access to the presentation: LINK

Albrecht Wirthmann (Eurostat) presented the current and future activities of Eurostat in the transition from Big Data to Trusted Smart Statistics (TSS). He discussed the organizational issues of the ESSnet Big Data 2018-2020 project, including the number of persons involved in particular research topics. He presented the TSS Strategy and the Roadmap of its implementation. It was pointed out that the overriding goal of all Big Data projects is to supplement statistical production with new data sources and to link them with traditional statistics. The significance of the TSS which are based on transparency, data access and management was emphasized.

Access to the presentation: LINK

Conclusions from the discussion:

- Eurostat plans to develop several threads in the future: Web Intelligence Hub, Trusted Smart Surveys, Mobile Network Operators data ect. (Albrecht Wirthmann)
- currently, the main focus is on the implementation of statistical production solutions developed under ESSnet Big Data 2018-2020. The key issue is the integration of Big Data with data collected by official statistics in a traditional way (Albrecht Wirthmann, Peter Struijs);
- it must be taken into account that it will not be possible to implement the developed solutions in all EU countries (due to *inter alia* legal and technological limitations and shortages of specialized staff) (Albrecht Wirthmann, Peter Struijs);
- the selection of data sources in the Big Data 2018-2020 project was determined by their availability and potential for the implementation in statistical production process (Peter Struijs);
- at the current stage, most of the activities initiated under ESSnet Big Data 2018-2020 are of experimental nature (Albrecht Wirthmann).

Recording available on YouTube.

Session 2 - WPF Process and architecture

(14:00-16:00 CET) / Discussant: **Peter Struijs** (CBS, Statistics Netherlands, NL)

Monica Scannapieco (ISTAT, Statistics Italy, IT) WPF leader, introduced the team of WPF and presented the assumptions of Big Data Reference Architecture and Layers (BREAL). BREAL was developed on existing UNECE and Eurostat architectural framework/standards: GSBPM (Generic Statistical Business Process Model), GAMSO (Generic Activity Model for Statistical Organisations), GSIM (Generic Statistical information Model) and EARF (ESS Enterprise Architecture Reference Framework). The elementary components of BREAL are the following layers: business, application, IT and technical. The BREAL architecture is used in the work of the following WPs: *Online Job Vacancies, Online Enterprise Characteristic, Smart Energy, Tracking Ships*.

Sónia Quaresma (INE, Statistics Portugal, PT) characterized the Business Layer of BREAL, including: principles, functions, its elements and relations between them. She stated that the principles were organized in categories relating to the big data life cycle phase in which they are applied/used (i.e. data capturing, data processing, relationship management). The following business functions were presented: acquisition and recording, data representation, data wrangling, enrichment of statistical register, integration of survey and register data, modelling and interpretation, validation, shaping the output, deployment, support of the statistical production, trust management and visual analysis. Apart from the main business functions, supporting functions were indicated: communication management, information management, knowledge and human resources management.

Bernhard Fischer (Destatis, Statistics Germany, DE) discussed the assumptions of the Application Layer, including the main and supporting services that relate to business parts and functions.

Tauno Tamm (Statistics Estonia, EE) described the BREAL's Information layer. The information layer consists of three sublayers: raw data sublayer, convergence sublayer and statistical sublayer. The information layer is based on the existing architecture concepts of data, metadata and GSIM.

Remco Paulussen (CBS, Statistics Netherlands, NL) discussed the BREAL operational model. He indicated that the main goal of the model was developing solutions architectures supporting knowledge sharing in the following dimensions: national, European and international. He emphasized that the WP has attempted to build common assets within the ESS. The WP focused on three types of sharing: application services, infrastructure platform, data sources. In order to create the solutions for sharing application services, ESS Enterprise Architecture Reference Framework (EARF) was adopted.

Collective presentation: LINK

Conclusions from the discussion:

- the work that has been done by the WP has the potential to be used in statistical and non-statistical organizations (Remco Paulussen based on the survey results, Peter Struijs);
- the material and documents developed within the WP should be treated as reflections, promoting certain solutions. These are not fully mature recommendations. It is a reflection on the work developed by the implementation groups (Monica Scannapieco, Remco Paulussen);
- basic standards and recommendations for statistical offices include GSIM and GAMSO models (Monica Scannapieco).

Recording available on YouTube.

Tuesday, 24 November 2020

Session 3 - WPB Online job vacancies

(10:00-12:00 CET) / Discussant: **Monica Scannapieco** (ISTAT, Statistics Italy, IT)

Tomaž Špeh (SURS, Statistics Slovenia, SI) the leader of WPB, presented the organizational issues of the work of the WP, including its goals and assumptions. WPB focused on using Online Job Advertisements data (OJA) and their integration into official statistics. The issues addressed by the WP involved *inter alia* the definition of statistical products, the concept of production processes at the national and European levels, and the development of experimental statistics from OJA. The OJA data appears to have a great potential to provide up-to-date real-time information about changes in the labour market, and can complement and enrich traditional statistical data. OJA data provides very detailed information about job characteristics (available daily, weekly and monthly). Due to the nature of the data, they can be used to detect and monitor labour market concentrations, as well as develop forecasts on OJV and flash estimates of economic indicators across the EU. The challenges tackled by the WP related to the use of OJA data in official statistics, including such issues as understanding how OJA are linked to actual job vacancies (JV).

Presentation: LINK

Francis Saucy (FSO, Statistics Switzerland, CH) presented the CEDEFOP data and the possibility of their use by official statistics. He concluded that OJA data offer a basis for trustful comparisons among countries but a comparability over time of OJA is not yet known.

Presentation: LINK

Tomaž Špeh (SURS, Statistics Slovenia, SI) presented a comparison of the data on job vacancies collected to date by the NSI with the data on OJA.

Presentation: LINK

Jakob de Lazzer (Destatis, Statistics Germany, DE) discussed the issue of labour market concentration (LMC). He noted that no published measures have been available on this subject so far, and presented an example of using OJA data from Germany to measure LMC index.

Presentation: LINK

Črt Grahonja (SURS, Statistics Slovenia, SI) discussed the use of OJA data for estimating economic indicators, including quick estimates of GDP and admitted that the results were not sufficiently satisfactory due to short time series and long lag.

Presentation: LINK

Tomaž Špeh (SURS, Statistics Slovenia, SI) referred to the use of OJA for Covid-19-related research and other studies in which OJA data are compared with some indicators (e.g. educational level, experience years). He pointed out that the greatest advantage of OJA data is that it can be presented in real time.

Presentation: LINK

Conclusions from the discussion:

- The challenge for using OJA data is its integration with statistical data (Monica Scannapieco);
- the study did not attempt to compare permanent and temporary jobs to OJV. The subject of the study was not vacancies but job advertisements (Tomaž Špeh);
- the challenge was the quality of the data (duplicates) and classifications using CEDEFOP data (the lack of stability of data and their variability over time) (Tomaž Špeh, Jakob de Lazzer);
- the problem with using this type of data is their stability as a data source (Francis Saucy).

Recording available on YouTube.

Session 4 - WPC Enterprise characteristics

(14:00-16:00 CET) / Discussant: **Monica Scannapieco** (ISTAT, Statistics Italy, IT)

Galya Stateva (NSI, Statistics Bulgaria, BG) the leader of WPC, introduced the team, discussed the objectives, tasks and main results of work. She indicated three main goals of WPC: improvement or updating of existing information (SBR); maximizing the quality and quantity of statistical results (ICT survey); ensuring the possibility of sharing resources at the ESS level. The tasks of WP included: designing and developing transparent rules for acquiring data from websites; defining the reference methodological framework (RMF); developing a starter kit for web scraping on enterprise characteristics; developing a quality template for web scraped data; producing experimental statistical products using collected enterprise data as the primary source. She listed the main results of the activities undertaken by the WP, including: the calculation of statistical indicators specified in RMF 1.0, integration of OBEC information at macro and/or micro level, methodological notes (explanation of how the results were obtained), reference metadata (reports on quality ESQRS and ESMS), the experimental statistics on the project website.

Presentation: LINK

Aidan Condron (CSO, Statistics Ireland, IE) described the assumptions of the basic technique of data collection – web scraping. He defined Web Scraping (WS) as web robots or "bots" replicating the human activity of finding web pages and capturing data. He drew attention to the principles and practices of web scraping, including: compliance with the website scraping policies being in line with the statistical principles set out in Regulation 223/2009 on European statistics; compliance with all applicable EU and national regulations; protection of personal data in accordance with the GDPR; maintaining transparency of the tools, methods and processes used.

Presentation: LINK

Olav ten Bosch (CBS, Statistics Netherlands, NL) highlighted functional production prototypes which have been made for two use cases: URLs retrieval and ICT usage in enterprise survey. Both new use cases added to NSI's data improvements, new indicators on enterprise characteristics and enabled validation or replacement of ICT

survey answers. A big data processing cycle for these two cases was designed and described in the Reference Methodological Framework (RMF).

Presentation: LINK

Donato Summa (ISTAT, Statistics Italy, IT) referred to BREAL presented application, information and solution architecture of OBECs's use cases (URLs Inventory of Enterprises and Variables in the ICT usage in enterprises survey).

Presentation: LINK

Jacek Maślankowski (GUS, Statistics Poland, PL) presented the Starter Kit for NSIs. He referred to applications: URLs Finder, Social Media Profiles, URL Scraper. The starter kit includes source code (in Python), description and step-by-step manuals.

Presentation: LINK

Heidi Kühnemann (Statistics Hesse, DE) presented web scraping results for Amazon sellers. The main goal of this case study was to test the possibility of combining scraped Amazon data with SBR using VAT and other registration numbers, but optionally also addresses and phone numbers.

Presentation: LINK

Conclusions from the discussion:

- the main problem of that sort of data is their integration with statistical data (Monica Scannapieco);
- it was noted that the data on Enterprise characteristics are presented by NSO as experimental statistics (Galya Stateva);
- the challenge for the further development of research on the issue is the lack of access to some websites. Hence, a change in legal regulations is required (Olav ten Bosch);
- additional challenges refer to informing website owners about the research to be carried out with the use of their data due to the lack of national guidelines in this matter (Galya Stateva, Olav ten Bosch);
- incorrect certificates confirming website compliance and errors related to blocking access to websites for web robots might pose yet another challenge (Jacek Maślankowski).

Recording available on YouTube

Wednesday, 25 November 2020

Session 5 - WPD Smart energy

(10:00-12:00 CET) / **Discussant: Piet Daas (CBS, Statistics Netherlands, NL)**

Arko Kesküla (Statistics Estonia, EE) made an overview of WPD goals, presented the definition of smart electricity meters and described the data storage system. Smart meter is usually an electronic device that records electricity consumption at intervals of an hour or less, and transmits this information at least once a day back to the tool for monitoring and billing. Smart grids are energy grids that can automatically monitor energy flows and adapt accordingly to changes in energy supply and demand. Combined with smart metering systems, smart grids reach consumers and suppliers by providing real-time consumption information. The following areas were addressed in the work of the WP: processing and analysis of data from smart meters; development of procedures, technical solutions for data from smart meters; developing functional production prototypes (statistics of electricity consumption of enterprises and households, identification of energy consumption patterns). Description of data processing from the preparation stage was presented, i.e. raw data processing, its combination with administrative data, classification of intelligent sensors (finding end users), to data validation and marginal data reception.

Presentation: LINK

Johan Fosen (SSB, Norwegian Central Statistical Office, NO) presented survey results for monthly electricity consumption based on the data obtained from smart meters. Its aim was to investigate how electricity data can help identify vacant dwellings. He pointed out that the study require taking into account many elements, e.g. the time of the measurement, the address of the smart meter at the apartment level, the access to the daily or hourly consumption of the smart meter and other information about the "signs of life" than the consumption (e.g. location of the mobile phone).

Presentation: LINK

Conclusions from the discussion:

- open issues that require further research involve: data granularity, data privacy (in the case of the low level of aggregation) and cooperation with data providers (Piet Daas);
- methodological solutions developed within the WP may be applied in other countries (Piet Daas, Arko Kesküla, Johan Fosen);
- access to data may be a problem – a contact with the data administrator is required (Arko Kesküla);
- data from smart meters is structured and easy to process (Arko Kesküla).

Recording available on YouTube.

Session 6 - WPE Tracking ships

(14:00-16:00 CET) / Discussant: **Piet Daas** (CBS, Statistics Netherlands, NL)

Remco Paulussen (CBS, Statistics Netherlands, NL) the leader of WPE presented the team and the agenda of the presentation.

Tessa de Wit (CBS, Statistics Netherlands, NL) – made an introduction to the Automatic Identification System (AIS). She presented AIS, including the types of information transferred, data sources and platforms that use AIS.

Presentation: LINK

Mark van der Spoel and **Tessa de Wit** (CBS, Statistics Netherlands, NL) presented the use of (AIS) in the monitoring of inland waterway transport on the example of Dutch waterways.

Presentation: LINK

Michał Bis and **Anna Bilska** (GUS, Statistics Poland, PL) presented the possibilities of using AIS in research in the domain of maritime economy. The aim of the WP's work was to develop a functional prototype to promote and support the AIS-based big data collection, processing and analysis, as well as development and implementation of the process of traffic and activities related to the sea fishing fleet. For subsequent work, they proposed the acquisition and testing of a new data source - Vessel Monitoring System (VMS) to generate experimental statistics using the developed process of marine fishing fleet traffic and activity.

Presentation: LINK

Eleni Bisioti and **Christina Pierrakou** (ELSTAT, Statistics Greece, EL) delivered a presentation about the use of the AIS data in the case of port visits in Piraues.

Presentation: LINK

Wai King Meijer-Cheung, Sjoerd Schenau and **Tessa de Wit** (CBS, Statistics Netherlands, NL) presented the possibility of assessing the energy consumption and air emissions of shipping vessels for the purpose of environmental accounts. The case study covered shipping vessels related to the Dutch economy.

Presentation: LINK

Conclusions from the discussion:

- AIS as a data source is applicable in a global perspective. From the point of view of statistical research, it can be used in currently conducted studies and new research directions (e.g. air pollution emissions) (Piet Daas);
- "fake records" and distance evaluation may generate some problems. However, all deviations from the norm are replaced with data from algorithms or are removed (Michał Bis);
- the analyses with the use of the AIS data can be conducted via the UN Global Platform.

Recording available on YouTube.

Thursday, 26 November 2020

Session 7 - WPK Methodology and quality

(10:00-12:00 CET) / Discussant: **Peter Struijs** (CBS, Statistics Netherlands, NL)

Alexander Kowarik (STATA, Statistics Austria, AT) discussed the work of WPK which consolidated the knowledge and know-how acquired from all work packages. He listed three synchronized work strands: typification, methodology and quality. In addition, a brief overview of the Quality Guidelines for the Acquisition and Usage of Big Data (QGBD) was made.

Presentation: LINK

Sónia Quaresma (INE, Statistics Portugal, PT) focused on Typification Matrix and Evolution Roadmap. She described the possibility of the usage of the Typification Matrix, i.e. the assessment of big data sources available in other WPs and their relationship with the qualitative guidelines for the acquisition and use of big data. The assessment maturity levels were based on the feedback from other WPs.

Presentation: LINK

Jacek Maślankowski (GUS, Statistics Poland, PL) presented the results of a literature review on the subject matter and online database with relevant research papers and their categorization.

Presentation: LINK

Piet Daas (CBS, Statistics Netherlands, NL) made an overview of big data methodology issues. He pointed out that currently a large part of statistical methodology is based on surveys and administrative data. In his opinion, big data is a by-product of modern society. There is still little knowledge of the big data generation process and of the units it contains. Working in a data-driven way is new for NSIs, therefore it is crucial to work on methods and principles to ensure that valid conclusions are drawn when using big data. Additionally, the examples of statistical use of big data sources were presented, followed by an overview of four steps of big data-based statistics production.

Presentation: LINK

Giovanna Brancato (ISTAT, Statistics Italy, IT) presented the results of the WP's work on the topic of Quality Guidelines including the list of common features and differences between the QGBD and Komuso Guidelines. She also presented pros and cons of the integration of these two manuals into a new one and maintaining them separately.

Presentation: LINK

Conclusions from the discussion:

- The added value of WP work is that it gathers experience from many NSOs and has made an extensive literature review. The challenge was the large number of new data sources that had to be analyzed (Peter Stuijs);
- the quality of data provided by private institutions remains a challenge (Alexander Kowarik);
- the task of the NSIs is to support civil data science. In statistical research, the information needs of local communities should also be taken into account (Sónia Quaresma, Peter Strujis).

Recording available on YouTube.

Session 8 - WPG Financial transaction data

(14:00-16:00 CET) / Discussant: **Marc Debusschere** (Statbel, Statistics Belgium, BE)

Johan Fosen (SSB, Statistics Norway, NO) delivered a presentation on the financial transaction data (FTD), including: definitions, data access possibilities, assessment of the potential for using FTD in official statistics and case studies. The analysis performed by the WP included financial transactions (money transfer from payer to payee) using various payment instruments and payment platforms. Data on payment transactions allows for the assessment of the scale of e-commerce, estimation of the early macroeconomic indicators, sharing economy (e.g. accommodation), household budget statistics, retail sales statistics, interactions between (business) sectors. It was pointed out that processing of FTD requires: the statistical act or formal agreement, ensuring that GDPR requirements are met, allocation of sufficient NSI resources, identification of relevant data providers, and guaranteeing secure data handling by the NSI. A general discussion on FTD specificities was followed by a presentation of a case study on retail sales statistics. Its goal was verification whether the FTD could replace current data sources for the retail sales statistics.

Presentations: LINK 1, LINK 2

Alessandra Righi (ISTAT, Statistics Italy, IT) presented a case study of macroeconomic forecasts for Italy. The purpose of this case study was to forecast private consumption, monthly retail and e-commerce sales before and during COVID -19 for the aggregated FTD time series. FTDs proved useful in macroeconomic analysis due to the high level of details and timeliness (data close to real time). However, it was noted that challenges for usage FTD in the statistical production are still very important and are related to data collection, validation, storage, interpretation and remaining privacy.

Presentation: LINK

Pedro Campos (INE, Statistics Portugal, PT) stressed that the main goals of WPG was to get acquainted with the sources and infrastructure of FTD data in the participating countries. In the case of the infrastructure, the main goal of the WP was to assess the statistical potential of this data source. In the case study carried out by Portugal, the data from the tax office provided by the Portuguese Ministry of Finance was used. On their basis, an assessment of the demand in two sectors of the economy was made: small accommodation services and car riding services.

Presentation: LINK

Giulio Perani (ISTAT, Statistics Italy, IT) addressed online transactions. E-commerce / e-business involve transferring data and money online in order to execute transactions. In order to measure them, appropriate "statistical units" have to be identified. In the case of E-business and FTD most purchases of goods and services online are paid for with credit cards (and e-money). A lot of purchasing information is available in the credit card's transaction data. Additionally, the issues of Shared Economy Platforms (SEP) and the possibility of using FTD by official statistics was raised.

Presentation: LINK

Conclusions from the discussion:

- this type of data has a wide range of applications in statistical research, including: tourism, accommodation, household budget surveys; however the main problem is access to data (Marc Debusschere);
- the interest in the topic was increased by Covid-19 (Marc Debusschere, Pedro Campos);
- the key issue is to build strong cooperation between NSIs, the Central Bank and national financial organizations in which the Central Bank will play an active role (Giulio Perani, Alessandra Righi).

Recording available on YouTube.

Friday, 27 November 2020

Session 9 - WPI Mobile network data

(10:00-12:00 CET) / Discussant: **Alexander Kowarik** (STATA, Statistics Austria, AT)

David Salgado (INE, Statistics Spain, ES) presented the main goal of the WP's work, i.e. the development of a reference methodological framework for mobile network data. He stressed out that the access to data from mobile operators is significantly limited. He also presented the results of the use case concerning an estimation of population in France, carried out in the Orange Labs.

Milena Suarez-Castillo (Insee, Statistics France, FR) pointed out the main challenges of the use of the mobile network operators' data in an official statistics: old national telecommunications regulations, perceived identification risks and personal data protection. In order to eliminate these limitations there is a strong need for the new e-privacy regulation that gives access only to anonymous aggregated data for statistical and scientific purposes; as well as obtaining data at the local level and putting more attention and research into statistical disclosure control.

Sandra Hadam (Destatis, Statistics Germany, DE) presented a few examples of using mobile network data from operators in Germany.

Tiziana Tuoto (ISTAT, Statistics Italy, IT) shared the results of the cooperation with mobile network operator in Italy for the purposes of preparing the Origin-Destination matrix for commuters and presence and movements of tourists.

May Offermans (CBS, Statistics Netherlands, NL) presented new issues and challenges with data-access by example of the Netherlands. He also drew attention to legal issues, anonymization of communication data, their aggregation and their local nature.

Bogdan Oancea (INS, Statistics Romania, RO) shared his experience regarding establishing cooperation with mobile network operators, underscoring the difficulties it entails.

Martinez-Vidal (INE, Statistics Spain, ES) and **Martijn Tennekes** (CBS, Statistics Netherlands, NL) explained how WPI implemented the idea to the work of WP. They also presented used software tools.

Collective presentation: LINK

Conclusions from the discussion:

- in the case of mobile network data, there is no permanent access to data for the purposes of statistical production (Bogdan Oancea);
- it is essential to develop European regulations related to the acquisition and sharing of data (Fabio Ricciato);
- WPI undertook to develop methodological solutions, libraries and programming solutions which can be applied to other research using mobile network data (Fabio Ricciato);
- the use of mobile network data in official statistics evokes negative social emotions. The development of a communication strategy on the use of mobile network data might mitigate the problem (Fabio Ricciato, Alexander Kowarik);

- the key is to maintain a balance between the information obtained and the protection of privacy (Fabio Ricciato).

Recording available on YouTube.

Session 10 - WPH Earth observation

(14:00-16:00 CET) / Discussant: **Alexander Kowarik** (STATA, Statistics Austria, AT)

Marek Morze (GUS, Statistics Poland, PL) WPH leader, discussed the use of the Earth observation (EO) data in official statistical production, in the context of improvement of the statistical registers, areal statistics and the upcoming 2021 Census and Agricultural Census in Poland. He also indicated the usefulness and practical application of Earth Observation data in order to fill in the gap between statistical and geographical information called "geospatial breakdown". Moreover, thematic tasks and case studies conducted within WPH were presented.

Presentation: LINK

Magdalena Mleczko (GUS, Statistics Poland, PL) presented an architecture framework for data using satellite imagery for the production of statistics on: agriculture, built-up areas, land cover and forestry. She indicated that the common denominator of the case studies analysed within the WP were the use of raw EO data to compute statistical ratios by applying image classification with machine learning algorithms. The BREAL operating model was used in the study. As a summary, it was pointed out that satellite data, depending on the scale and continuity (time) of measurements, indicate high complementarity with traditional statistical methods and constitute a supplementary source of in-situ correctness verification. They allow direct calculation of indicators and visualization of the geographic dimension and context of the indicators. EO data is free and open source.

Presentation: LINK

Przemysław Slesiński (GUS, Statistics Poland, PL) presented the results of a case study on the use of Sentinel 1 and Sentinel-2 satellites for mapping and estimating agricultural crops. He presented the various stages of satellite data processing in order to obtain statistical information. The results of the work indicate the enormous potential of the EO data for official statistics due to an easy access to the data, size of the research area (from small to global scale), reliable results (high accuracy based on group of crops), and the reduction of statistical production costs. On the other hand, the existing limitations involve complex processing, low spatial resolution (10m) which can eliminate small parcels, terrain influence and the small differentiation between certain crops due to the similarity in plant structure.

Presentation: LINK

Maria Yli-Heikkilä (Luke Institute of Natural Resources, FI) presented the results of a case study on the monitoring of the off-season vegetation coverage. The study showed that classification outcome is very promising and the soil cover indicator can be published even within 7 days from the latest Sentinel image acquisition.

Presentation: LINK

Vianney Costemalle (Formerly Insee, Statistics France, FR) presented the results of the work related to the case study on the implementation of the SDG 11.7.1 indicator, 11: Sustainable cities and communities based on satellite data. He described the methodology issues, sensitivity and quality analyses, as well as the results of work involving satellites imageries, including open public spaces and build-up areas.

Presentation: LINK

Magdalena Mleczko (GUS, Statistics Poland, PL) discussed a case study on the characteristics of the urban sprawl across urban areas in the Netherlands with data-based machine learning methods.

Presentation: LINK

Maren Köhlmann (Destatis, Statistics Germany, DE) presented the results of the case study "Urban quality of life" focused on the urban heat islands. The study was based on the data from the micro census and plant area. It proved that the integration of earth observation data and geocoded official statistics have the potential to answer new research questions and be used to elaborate policy recommendations on a relatively deep regional level.

Presentation: LINK

Bertrand Ballet (SSP, Ministry of Agriculture and Food, Department of Statistics, France) shared the results of a case study which compared "in situ" and "remote sensing" data collection methods for land cover data.

Presentation: LINK

Diego Zardetto (ISTAT, Statistics Italy, IT) presented a case study on the development of a prototype software system which employs deep learning algorithms (CNN + U-Net) to extract data from satellite imageries and based on them, automatically estimate the coverage terrain.

Presentation: LINK

Rossano Figueiredo (INE, Statistics Portugal, PT) discussed a case study which embraced of two topics. The first one involved an update on the INSPIRE Theme Statistical Units dataset, namely the Settlements and Enumeration Areas. The process would contribute to building the geospatial framework to support 2021 Census. The second one explored the possibility of studying forest and the eucalyptus plantations and its impact on preventing forest fire.

Presentation: LINK

Conclusions from the discussion:

- the added value of the project is the development of very good training sets for modelling, classification (Diego Zardetto, Bertrand Ballet)**;**
- the challenge of using satellite data is its integration with traditional data from statistical surveys.

Recording available on YouTube.

Monday, 30 November 2020

Session 11 - WPJ Innovative tourism statistics

(10:00-12:00 CET) / Discussant: **Marc Debusschere** (Statbel, Statistics Belgium BE)

Marek Cierpiał-Wolan (GUS, Statistics Poland, PL) WPJ leader, presented an overarching goal of the WP, i.e. the development of a conceptual framework and a prototype of the Tourist Information and Monitoring System. The work of the WP resulted in the production of tourism statistics by integrating various big data sources, administrative registers and statistical databases with the application of advanced statistical methods. Specific goals of the work undertaken involved the identification and assessment of the availability and quality of big data sources; the development of the methods of combining and temporal disaggregation of tourism data; providing recommendations on the creation of high-quality data for tourism statistics; and setting up a smart prototype of Tourism Integration and Monitoring System with the identified data sources.

Galya Stateva (NSI, Statistics Bulgaria, BG), presented the results of work on the implementation of scraping methods to obtain data from websites (e.g. hotels.com, booking.com, pochivka.bg).

Christina Pierrakou (ELSTAT, Hellenic Statistical Authority, GRE), presented the results of works on the development of methodology for combining and aggregating data from various sources, including web scraping data.

Mascia Di Torrice (ISTAT, Statistics Italy, IT), presented inventory of new data sources (identification of the most popular websites in the tourist sector, selection of data sources that can be useful in touristic statistic). She also presented methods of comparing the regional archive with the results of web scraping.

Shirley Ortega-Azurduy (CBS, Statistics Netherlands, NL) presented the results of usage of web scraped data for the construction of the Accommodation Enterprises Database and preparation of flash estimates on the number of tourists by hotels.

Rui Alves (INE, Statistics Portugal, PT), presented the results of the study in the field of geocoded web scraped data.

Boris Frankovic (SOSR, Statistical Office of the Slovak Republic, SK), shared his experiences with web scraping data and combining them with traditional data sources.

Collective presentation: LINK

Conclusions from the discussion:

- the challenge is the access to data from tourist websites and the cooperation with the private sector, (Marc Debusschere);
- the problem is also the integration of the data from different sources (Marc Debusschere);
- in order to implement the solutions developed within the WP it is essential to introduce appropriate legal regulations which will ensure constant access to data sources for the official statistics (Christophe Demunter);
- the development of methodological and IT solutions by the WP may be implemented at various territorial levels (Christopje Demunter, Marek Cierpiał-Wolan).

Recording available on YouTube.

Session 12 - WPL Preparing smart statistics

(14:00-16:00 CET) / Discussant: **Michail Skaliotis** (Ext-Eurostat)

Natalie Rosenski (Destatis, Statistics Germany, DE) introduced the team and listed the case studies carried out by WPL. The goals of the WP were exploration of the IoT in order to produce trusted smart statistics, an overview of new and relevant topics for official statistics, recommendations for possible follow up studies. The case studies presented involved: smart farming, smart cities, smart devices, smart traffic. Finally, the identified challenges, i.e. data standardization and a lack of clear solutions were discussed.

Alexander Kowarik (STATA, Statistics Austria, AT) presented the results of a case study in the area of smart farming (Digital Farming). He pointed out that data sources in this field that could be used by the public policy include farm and livestock management systems. The challenges of the implementation of this type of data for statistical production are standardization of interfaces and low adoption rate of any specific solutions.

Georgi Hristov (NSI, Statistics Bulgaria, BG) presented the definition of smart cities and discussed the possibility of obtaining statistical data with the use of the Varna Limitless application which monitors the movement of different group of users.

Natalie Rosenski (Destatis, Statistics Germany, DE) presented the results of a case study in the area of smart city. The purpose of the study was to make an overview of "smart solutions" in different cities of the European Union and of the potential of their outputs for statistical production. As part of the study she explored 12 Smart Cities founded by the EU Research an Innovation Programme Horizon 2020 and the German Smart Cities Darmstadt and Berlin.

Marie-Pierre de Bellefon (Insee, Statistics France, FR) presented the results of a case study in the area of smart city, regarding the possibility of combining sensor data and socio-economic data for the purposes of characterizing people exposed to air pollution. In her study she used the example of Nice.

Olav ten Bosch (CBS, Statistics Netherlands, NL) presented the definition of a smart device, Eurostat reference model and accessible data from smart devices.

Pontus Lindroos, (Statistics Finland – Tilastokeskus, FI) presented the results of a case study in the field of smart traffic data for GDP early estimation. In the study the data from The Finnish Transport Infrastructure Agency and Traffic Management Finland were used.

Johan Fosen (SSB, Norwegian Central Statistical Office, NO) presented the results of work in the field of truck sensors for road freight transport. He discussed the potential of using the Remote Freight Management Systems in official statistics.

Collective presentation: LINK

Conclusions from the discussion:

- the challenge for a wider use of big data in official statistics is to connect the different data sources and to solve cross-cutting problems that arise (Michail Skaliotis);
- It is important to create a community of smart statistics that will grow independently of projects (Michail Skaliotis);
- the activities carried out under WPL serve to identify new topics for the purposes of official statistical production.

Recording available on YouTube,

Peter Struijs thanked all the participants, the panellists and Eurostat and closed the meeting.

Big Data for European Statistics (BDES 2021) Roundtable

16-18 June 2021

REPORT

BDES 2021 Roundtable was the ESSnet Big Data II project's closing event, organised online via Zoom on 16-18 of June 2021. The aim of the BDES 2021 Roundtable was open and free-ranging discussion on the future directions for big data and new data sources in official statistics by a small group of invited experts. The event gathered 47 participants from 28 organisations, including national statistical offices, ministries and research institutes.

On the 16 and 17 of June 2021 the Roundtable was divided into two sessions, one in the morning (10:00-12:00 CEST) and one in the afternoon (14:00-16:00 CEST) of each day, and every session consisted of four parallel breakout groups with 5 to 13 participants discussing one of the eight themes. Attendees were assigned to a particular breakout group in line with the preference ranking and in a way to ensure their participation in four different theme discussions during the whole event.

On the 18 of June 2021 two plenary sessions (one in the morning and one in the afternoon) were held to report the results of all breakout group discussions and to further exchange ideas in the whole group. Each session covered four themes.

In order to guarantee free and open speech in the parallel sessions, the Chatham House Rule was implemented. According to this rule participants did not speak for an institution or in any formal role, but as individuals with personal ideas and views. Moreover, in the reports of the parallel sessions, individual contributions were not identified.

The Roundtable was officially opened by dr. Emanuele Baldacci (Director of methodology, IT and corporate statistical services, Eurostat, on behalf of Mariana Kotzeva, Director-General of Eurostat) and dr. Dominik Rozkrut (President of Statistics Poland). The sessions were mostly chaired by Marc Debusschere (Deputy Project Coordinator), and on Friday afternoon by Peter Struijs (Proejct Coordinator), and each breakout group had assigned a moderator who facilitated the exchange of opinions, and a reporter who summarized the conclusions from the discussion.

Following is a short description of 8 discussion themes, further elaborated in Annex 2:

1. **From exploration to exploitation** – the theme addressed the purpose of the implementation of new data sources in official statistics, the role of experimental statistics, and the roadmap for the integration of new data sources in the production process, with a special emphasis on the involvement of users on different stages.

2. **Leaner and meaner statistics** – the theme covered the question of how official statistics could response in a quick and more flexible way to users' needs and new emerging challenges, preserving the standard and quality of statistical outputs.

3. **Access modalities** – the theme included various aspects of the access and use of new data sources, including legal, infrastructure and methodological issues, knowledge and skill demand, but also diverse data access modalities for development and production stages.

4. **International and ESS cooperation and division of labour** – the track discussed the organisation of work within the ESS regarding different types of big data sources (AIS and MNOs, among others) and the possibility of coordination and cooperation in various dimensions.

5. **Fostering culture change** – the main questions covered by the topic involved identification of enablers and obstacles to culture change which may have an impact on the implementation of new data sources in statistical production.

6. **Smart statistics** – the theme addressed the issues related to the definition of smart statistics and trusted smart statistics, the roadmap of their development and the role of collaboration within ESS, between NSIs, and with external users.

7. **Business architecture and methodological frameworks: next steps** – the role and further development of the frameworks created within the ESSNet Big Data II projects' WP F and WP K were the main interest of this discussion track.

8. **Further research priorities** – the focus of the discussion was on strategy of prioritizing new data sources which take into account their potential for moving from the experimental to implementation phase, as well as users' needs.

Detailed summary of all breakout group discussions, and additional points raised during the plenary sessions can be found below.

Theme 1. From exploration to exploitation

Session: 16 June 2021, 10:00-12:00 CEST

Moderator: Sofie De Broe (CBS, Statistics Netherlands)

Reporter: Tomaž Špeh (SURS, Statistics Slovenia)

How to implement new data sources concretely in the production of official statistics

This discussion followed the issues proposed in the briefing notes:

- What main role do you see for new data sources in future regular statistical production?
- What is the best starting point for introducing new data sources?
- What sequence of implementation steps should best be followed?
- What is the role of experimental statistics?
- How do we take user needs into account?

Regarding the main role of new data sources in future regular statistical production the main points were:

- The main role of official statistics is to produce reliable, trusted and high quality indicators.
- New data sources provide opportunity to produce new indicators not provided by official statistic, faster, more granular, to facilitate necessary insights for changes in society.
- The role and legal mandate of official statistics should be addressed, how to convince production units.
- New data sources could supplement current data inputs and expensive data collection.
- New data source could be used as a complementary source to validate existing statistical products, considering statistical domains, type of data, quality dimensions.
- New data sources are available to all society, the role of official statistics is to educate and to raise awareness.

Regarding the best starting point for introducing new data sources the main ideas were:

- First, identify user needs, than identify data source(s) needed to provide the answer.
- Any data source that can facilitate this should be used, but not request more data than needed to answer the question.
- International project are good starting point since not all statistical institute can allocate resources need.
- Collaborate with researchers and private data owners (data ownership and data altruism).
- Always do both experiment and production.
- Be transparent, open data, open code, preserve privacy.
- Highlight success stories.

Regarding the sequence of implementation steps the following were discussed:

- New data sources needs to be repurposed in order to be used in official statistics. This takes time, development of complex procedures and investment are needed to move to production.
- Extend and adapt current rules and procedures of official statistics.
- Perform detailed data analysis and check long-term availability of the data source before complex data processing.
- Produce experimental statistics and provide support to people that believe, have passion and time to experiment, establish research laboratories, success stories needed.
- International projects are good starting point.
- Educate and raise awareness of the users, new players in the industry, explain challenges (coverage, statistical units,…).

Regarding the role of experimental statistics, the following were identified:

- To support the needs in changing society timely.
- To produce additional indicators, not provided by official statistics.
- To provide timely and more granular indicators.
- To open code, methods, new chapter to produce transparent statistics.
- To consult and engage users, promote citizens science.
- To collaborate with partners.

Regarding how to consulting user needs:

- There is a need to consult users' systematically and in a formal way.
- On approach is creation of statistical boards consisting of important users of official statistics, private data holders, researchers and academia.

Session: 17 June 2021, 10:00-12:00 CEST

Moderator: Elise Coudin (Insee, Statistics France)

Reporter: Albrecht Wirthmann (Eurostat)

The group was attended by 12 persons. The discussions of the group focused on 3 topics with the following conclusions

1. Role of new data sources

When assessing the role of new data sources or better the role of statistical data derived from new data sources a distinction between new and existing information needs should be made.

Use of new sources in relation with existing needs

Statistical information derived from new data sources is used in combination with traditional statistics. Most often, the additional information is used to supplement current information and add aspect of granularity in time, i.e. make information more timely or achieve higher frequency of statistics, in space, i.e. to produce estimates with higher spatial resolution, or for smaller territorial units, and to add information that would allow to make better distinction in the level of classifications. Sometimes, additional variables can be added that are not collected or collectable via traditional sources.

On the other hand, the quality of data from different sources (new and traditional) can be better assessed and validated putting them together.

Use of new data sources can lead to reducing burden and cost of existing surveys when the sample sizes could be reduced.

New data sources are secondary data sources. The collection of the data is not motivated by the purpose of using them for official statistics. They have to be re-purposed. Statistical concepts, classifications and definitions have to be reconsidered to enable use of these data in official statistics, e.g. definition of residence derived from mobile devices' data.

New information needs

Statistics derived from new data sources answering new information needs can be stand-alone. Surveys could support quality (accuracy) of estimates from new data sources, e.g. surveys producing information on the use of mobile phones during holidays, in the daily environment, to compensate for inherent bias or improve estimates from convenience samples.

In general stability of access to new data sources is very important for engaging in use of these data. Methodological approaches for compensating instability of sources is important.

2. What is the best starting point in using new data sources?

There was some disagreement in the group concerning this topic. Some persons put user needs upfront while others emphasized the interaction between data scientists and domain statistician for the development of new statistical products. The dynamic interaction between data scientists and domain experts was considered as important.

An opportunistic approach was brought forward, looking at activities of other stakeholders, within the statistical community or in interaction with academia. The need for statisticians to be pro-active was stressed, in the analysis of data and needs, drawing up of specification, enabling interaction between stakeholders (data scientists, data holders, statisticians and users). Entering into an intensive dialogue with users was considered as important. This dialogue should also be used to educate users (policy makers) as regards to quality expectations and requirements of official statistics.

The crisis brought an opportunity for exploring new data sources and integrating them into regular statistics. Use of new data sources in one area acts as catalyst for adopting the same sources for other statistical products.

3. What **sequence of implementation steps** should best be followed?

There is a clear distinction between the development and the production phase. In each phase different requirements have to be fulfilled, e.g. in the production phase, the issues of reliable access to data has to be solved.

Inclusion of stakeholders is seen as a way to overcome the blocking point from experimentation to regular production of statistics.

During the development phase methodological and subject matter department should closely work together. External experts should be included when necessary. A way for shortening the development phase is to search and examine already developed solutions and take stock of existing knowledge and competences. When examining existing developments or solutions, these solutions should be transparent, preferably open source to comply with the requirements of the code of practice. Instead of establishing a provider/client relationship statistical offices should establish partnerships with data holders, academic or other stakeholders to co-develop and avoid buying black box solutions. The issue of reliable access to data should be ensured and the possible instability of data sources should be assessed and mitigated. There is a distinction between experimental statistics and output of the research and development phase.

The production process should be carefully designed, the involved actors should be identified with assignment of responsibilities. Engagement should be created and ensured. During the early phase of the production, communication between involved parties is very important.

The overall effort of transferring experimental statistics to regular production should not be underestimated.

Theme 2. Leaner and meaner statistics

Session: 16 June 2021, 10:00-12:00 CEST

Moderator: Barteld Braaksma (CBS, Statistics Netherlands)

Reporter: Elise Coudin (Insee, Statistics France)

- Room to be leaner and meaner, especially for covering areas that are changing rapidly

There is room for NSIs to offer new statistics complementing the traditional ones, in a more agile way, especially for covering areas that are rapidly changing (e.g. climate change, mobility, sustainability, crisis, new phenomena), areas that demand to be more innovative (platform economy) and by using new technologies/new emerging data sources which have potentialities. Use the reference image of NSI as a strength.

- Dare to be innovative and proactive, show experimental results

NSIs should not wait for all the conditions to be fulfilled to experiment and produce new results. They should experiment, show and communicate on experimental results, being transparent on what is done in accordance with our code of practice.

-Active communication and dialogue with stakeholders to seek social license and identify user needs

NSIs should communicate more, especially instauring dialogues with users and stakeholders in order to understand and adapt to the needs of the users, but also to gain social license (public opinion acceptability) on what they do, and gain support of stakeholders.

-Be meaner to IT security boss

There is a challenge in articulating security requirements for regular production and innovation. Within NSIs the roles and decision power of domain statisticians, IT security staff and innovative staff should be well defined to foster innovative project implementation.

-Integrate domain experts within the innovation labs and vice versas

This relates to acculturation of both datascience world and domain expertise world. Domain experts (and IT decision makers) should see the benefits of an innovation to adopt it, e.g. R&D departments can be created welcoming domain experts (and IT staff too), or reversely data scientists/engineer/architect positions should be created in the domain/business units.

-Support network building - ease exchanges of experts

Eurostat should support network building by expanding programs of exchanges of experts. Gentleman agreements and direct collaborations should also be encouraged.

-Be more proactive in fundraising, adopt a benefit/cost approach, involve supports from outside

Both NSIs and Eurostat should be proactive in fundraising. NSIs should seek for supports from outside (e.g.ministries, other national authorities, universities, cities/local government, potential private investors in the domain) and engage users. They should adopt a speech that compare costs and benefits of their improvements.

-Need for longer, renewable grants; improve support for expanding statistical programs

NSIs expose their needs for longer, renewable grants and for support for expanding statistical programs, especially in preparation of the new regulation context for data currently setting up.

-Actively try to learn from each other, inside and outside the ESS - ESSnets are important!

Session: 17 June 2021, 10:00-12:00 CEST

Moderator: Fabio Ricciato (Eurostat)

Reporter: Sónia Quaresma (INE-PT, Statistics Portugal)

Current Status – some success stories, but more need to be done

Not everything is rigid in NSIs. Success stories show already some flexibility and capability to adapt (e.g. reaction to emergency situation with pandemic) and innovate, but much more needs to be done.

Organization & Orientation – improve understanding of user needs

NSIsand the ESS must improve their capability to identify the societal needs (including future and latent needs). More attention needs to be paid to engage end users' for identifying future statistical products and improve existing ones. NSIs can learn from the private sector how to engage end-users. Avoid (waiter Mode – just asking the user what they want; arrogant mode – we tell them what they need).

Collaboration Practices – open up

NSI should make an effort to "open up" and collaborate more systematically with researchers, data holders, other public agencies and between regions/countries. Opening up also allows us to pull knowledge and learn new practices.

Relation to Other Data Holders – learn from them, not follow them

NSI have the opportunity to learn from some data holders, but without being "followers". NSI should exercise critical view on which statistical products and data sources are most promising, without necessarily following ephemeral sexy trends in the private sector. Also, competition is not really in place as long as NSI deliver to the promise of "quality" (including transparency, impartiality, independence…) that is the key differentiator against private providers of statistical information.

Multisource

The goal of NSI should be to produce statistical information from multiple sources. It allows improving stability (contrast volatility of individual sources), quality, reliability and independency of statistical production. Multisource should be a differentiator of official statistics vis-à-vis private providers of commercial statistics.

Make Statistical Regulations more innovation-friendly

Regulations define our mandate but are also a bottleneck for innovation. **Current regulations are not innovation-friendly**. Future statistical regulations should be designed in order to promote innovation (both in experimentation and transition to production) and also provide adequate resources for it.

Agility and innovation are costly

Being agile and innovative is costly (but *not* being so is even more costly in the long run!). The current funding model is tied to strict and detailed regulations focused on *current* statistics. Regulations should enable funding for the development of future statistics.

Theme 3. Access modalities

Session: 16 June 2021, 10:00-12:00 CEST

Moderator: David Salgado (INE-ES, Statistics Spain)

Reporter: Monica Scannapieco (Istat, Statistics Italy)

Access Modalities were first discussed by referring to Figure 1, where the *degree of granularity of data* to be accessed (ranging from Nano to Statistics data) is used as a dimension with respect to which consider several issues, namely: IT resources, Privacy risk, Data integration utility, External production costs, Black box risk, Specific knowledge, Skill demand, Multiple-domain utility, Communication policy complexity, Provider multiplicity complexity, Process &quality complexity, Source time-continuity. In the Figure, as an example, "Privacy risk" is supposed to decrease when passing from nano data to statistics; instead, "Black-box risk" is supposed to increase with an higher granularity of accessed data.

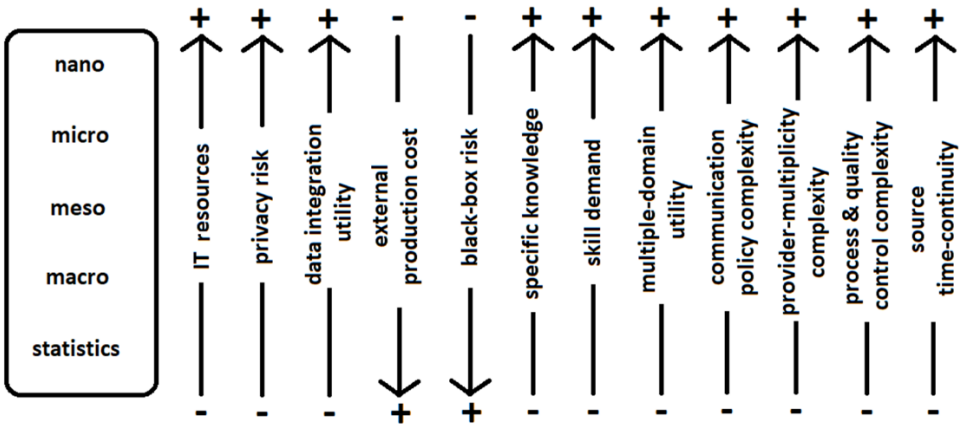

Figure 1: Data Access issue wrt Granularity of data

The overall discussion can be summarized along 5 main topics/themes, namely:

1. **Legal support to access Big data sources**. Here, all the group agreed on the fundamental need of a legal support stating: (i) a general law requirement, related to the right to have access to Big Data sources by NSIs for statistical purposes, and (ii) a specific law requirement in which a protocol between NSIs and Data Holders should be enabled by law by considering some important aspects including Provider Multiplicity, Source time-continuity, Black Box Risk External Production Costs, Privacy Risk, Involvement of private data holders in research.
2. **Research and production stages** to access Big data sources. The group agreed on a clear distinction between a research stage and a production stage. This distinction does have an impact on the degree of data granularity that should be higher (nano/micro) for research and can be lower (macro/aggregated) for production. Indeed, for research purposes, there can be the need to explore and set up methods and parameters that, at a later production stage, can be given for granted. Auditing mechanisms could also be set up in the production stage to permit access to lower aggregated data, if necessary.

3. **Peculiar characteristics of Big data sources.** Big data sources can be very different among each other and access to each source may require different approaches. Specific knowledge and skills requirements to process a Big data source can be necessary and the access solution should take that into account and lead to access modalities that consider pre-processing solutions that are not in-house.
4. **EU-level Investments.** The investments at EU level should be done on the legal framework for the access that was recognized by the group as the highest priority issue. In addition, infrastructure and resource sharing at EU level was also discussed to reduce the costs of National level investments.
5. **New models for collaborations with private data holders.** It was recognized that NSIs should engage specific partnerships with private that holders, which go beyond private-public partnerships, and are more focused on production of statistics for the public. This means that new business models should be set up, e.g. based on co-development for creation of public value. Probably, the ESS as a whole needs to be supported in identifying and adopting such models.

Session: 17 June 2021, 14:00-16:00 CEST

Moderator: Fabio Ricciato (Eurostat)

Reporter: Margus Tiru (Positium)

1. **Transparency is key**. Methodological transparency is a critical requirement and statistical authorities should not compromise when dealing with external data, as this is necessary to build trust (in the process, in the statistical output). Processing methods adopted for official statistics should be **open**, not proprietary. To this aim, processing methods are preferably developed by the statistical system or co-developed with external experts. They should be **reproducible** and **auditable**, and for this reason they need to be documented precisely along with **open-source code** reference implementations. If transparency is preserved, then processing the data at the premises of the data holder is likely the preferable solution in many cases (but not all).
2. **Transparency has a cost**. Development of good methodologies and good implementations thereof is an expensive process. But once methodology is developed, there are no major obstacles to running the methodology at data holders' premises (e.g., no obvious IPR problems were identified). While not low in the absolute sense, the cost of being methodologically transparent is lower than the ***cost of being not transparent*****,** (e.g., in terms of NSI becoming dependent on methodology providers, risk to lose public trust and credibility, etc..).
3. **Privacy and data decentralisation**. Data centralisation should be preferably avoided in order to (i) prevent risk concentration and (ii) prevent NSI being perceived as a gateway for government surveillance. When data across multiple data providers need to be combined before aggregation, data concentration can be avoided by resorting to computing infrastructures based on advanced Privacy Enhancing Technologies *that are designed to keep the data (or at least the control on the data) distributed*.
4. **EU level for cost-efficiency.** Pooling the process of methodology development and infrastructure design/development/procurement at EU level to divide (rather than multiply) the cost.

5. **Acquiring new skills**. Individual NSI and the ESS as a whole need to acquire new skills and learn to do things they are no (yet) good at. One example: developing and maintaining open-source code for complex software projects. Domain-specific knowledge is needed to develop good methodologies (e.g., telco experts for mobile phone data, image processing experts for satellite data). The process of "acquiring new skills" should pragmatically consider a combination of different channels, from training existing staff, to recruiting new staff, to collaborating/partnering with external experts or even outsourcing to external specialists specific tasks.
6. **Differentiate data access modalities for development and production**. Direct access to limited samples of raw (or quasi-raw) data by NSI staff is necessary in the process of methodological development and to familiarise with the characteristics and challenges of the new data. In a large-scale regular production, direct access is not necessary and "access to raw data" should be intended in the sense that the processing methods/algorithms defined (or co-defined) by the NSI/ESS run as close as possible to the system that generates the data.
7. **(shared) infrastructures**. The statistical system should expand IT infrastructure capabilities. Again, cost effectiveness considerations push towards infrastructure sharing. It was noted during the discussion that sharing infrastructures across NSI of different countries, i.e., at the ESS level, might be preferable than, say, sharing infrastructure between one NSI and another public agency in the same country (e.g., public national cloud). So again, shared ESS infrastructures appear to be the preferable way to go.
8. **Legal framework enabling flexibility**. The EU regulatory framework on accessing privately held data for official statistics should provide general rules and common principles. They should leave flexibility for arranging different detailed access models between the statistical authority (at national or EU level) and the data holder.

Theme 4. International and ESS cooperation and division of labour

Session: 16 June 2021, 10:00-12:00 CEST

Moderator: Albrecht Wirthmann (Eurostat)

Reporter: Peter Struijs (CBS, Statistics Netherlands)

The group session was attended by nine persons. The group focused on five topics and reached the following conclusions:

1. It is preferable for new data sources to be **accessed directly at the international level** if they fulfil certain conditions; these need to be specified.

There are several important considerations. Direct international access may have advantages if the data is highly standardized, the data source has a high territorial coverage, cross-border data needs can be served, the data holder operates cross-border or internationally and is organized correspondingly, and it results in data being collected only once. Small countries that do not have the capacity to invest in new data sources may benefit. If these characteristics are absent, the data is language-specific, or relevant legislation is highly country-specific, the disadvantages may dominate. Quality should be anyway sufficient, this may need to be researched in a pilot. For instance, AIS data is highly standardized, and MNO data is in some respects. Web-data is not standardized and is often language-specific. Direct international access may take place at EU level, but in some cases the world level can be considered. Mixed organization might be conceivable in order to tap advantages in different aspects, e.g. collect data from international platforms and organize processing in language paths. This may require integration of the production process.

2. The ESS should be open to **common data processing** in certain situations, which need to be defined.

Shared data processing can be considered if certain conditions are met, such as that quality can be managed, continuity can be ensured, necessary training can be arranged, etc. However, integration of shared processes in the national context may be a considerable hurdle to take. The best driver for common data processing would be output requirements, such as on coherence. Mechanisms for common data processing will be needed, and facilities such as platforms. Openness and transparency should be the norm, regarding e.g. methods and code. The ownership of the process must be clear, and responsibilities shared. Helping each other would be a condition, and ideally there is a network or community to fall back on. Furthermore, mixed models might be conceivable, e.g. joint processing of data for NSIs not having adequate capacities and individual processing for those who have the capacities.

3. Even in cases where no legislation is considered at the EU level, there is a case for **tuning national legislative initiatives** that have an impact on official statistics.

There is an interaction between international and national legislation, but this interaction is not symmetrical: international legislation is a joint affair, but national legislation is not tuned or harmonized unless there are legal international constraints. This situation should be changed, especially in the light of the changing role of NSIs. Also the survey on national legal obstacles to data access, which was held by the PG TF, showed that NSIs operate in hugely different circumstances. The ESS would have been in a different position, had the legal situation of EU countries been better tuned.

4. The international **division of labour should be based on principles** that need to be specified.

National statistical systems have structural differences, but may also be at a different stage of development. An example of the former is the choice of some countries not to have a population register, an example of the latter is lagging access to administrative registers for statistical purposes. The question what arrangements are desirable at the international level, and the path toward such provisions depend on this distinction. Also, international solutions should fulfil certain conditions, such as on funding and open and transparent decision making. Shifting the division of labour towards the international level should not reduce flexibility in responding to user needs.

5. The international statistical **educational and training programs** need to be reassessed, further developed and more strongly coordinated.

There are valuable training and education programs, such as ESTP courses and EMOS. It is important to keep them up to date, given the rapidly changing demand. New needs and expected future demands have to be considered and essential competences identified. New ways of delivering trainings should be explored to maximize the impact on the ESS. Collaboration with academia and universities could be conceived to offer specific courses for staff of statistical offices, e.g. master in data science.

Session: 17 June 2021, 14:00-16:00 CEST

Moderator: Sónia Quaresma (INE-PT, Statistics Portugal)

Reporter: Olav ten Bosch (CBS, Statistics Netherlands)

Division of labour:

Who can or should coordinate all or part of this:

- ESS WGs should coordinate the work on ESS level and also discuss ESSnets. Eurostat has a supporting and encouraging role. Staff exchanges, also among NSIs, and COE's are good instruments to improve cooperation.

The ESS is a coalition of the willing. Hence the member states represented in ESS working groups should coordinate the work on ESS level. Eurostat has a supporting and encouraging (pushing, not enforcing) role and can act on behalf of the WGs if instructed to do so. Also the launch of new ESSnets should be discussed in these WGs. Staff exchanges – also among NSIs - and centres of excellence are good instruments to improve efficient cooperation.

Subsidiarity within ESS:

How to organise the work:

- No NSI can do the work alone. International cooperation is needed to exploit the growing number of new global data sources using collaborative knowledge in a standardised way.

During the last decade the data landscape quickly changed from a ***data scarce***- to a ***data-rich*** world. No single NSI can do the work that has to be executed alone. The quantity of new data and data sources is overwhelming. Many new data sources are themselves ***international***. In some cases they are in the hands of big firms which operate ***beyond national***

borders. These sources cannot be acquired nationally. The knowledge required to efficiently and effectively interpret and process these data sources can only be managed collaboratively. Although it's an illusion to make generic solutions for all new data sources, from a standardisation viewpoint it is also inevitable to join forces. This is not always the case now. Example: national web scraping solutions. Fortunately there are also successful examples of cooperation like the web scraping policy template.

- Successful international cooperation cannot be done without the involvement of subject matter specialists in the right phase, both at international institutes as well within NSIs.

For efficient international cooperation it is necessary to be flexible and open towards solutions that might work less optimal in the local context but could be profitable on a global scale. It is necessary to align views and developments inside NSIs between innovation and subject matter specialists and between these groups at international institutes. Sometimes the exploratory phase needs less involvement of subject domains at the start, but it is important to go back to specialists at the right time, otherwise there is the risk of sending out contradictory signals. (Example: WIH).

Data subsidiary:

- Global data can and should in some cases be collected/coordinated centrally. Example: AIS. The best way, from efficiency, subsidiarity and user perspective viewpoint should be examined on a case by case base. International organisations should better align their initiatives.

Global data sources that cover multiple countries are likely candidates to be collected and coordinated centrally. The AIS data on the UN Global platform is a successful example. There is a risk of NSIs feeling their role reduced. Spreading the work and joint innovation are ways to reduce the risk. Many international organisations have their own (sometimes competing) data initiatives. There is a need for better alignment. Targeted international organisations seem to be able to move faster than others. Should the ESS get meaner and leaner? Innovation should not start from regulations but from user needs. Budgeting systems sometimes hinder innovation.

Processing subsidiarity:

- Processing can be done centrally where useful and feasible. The responsibility for processing national statistics stays at national level. National data needed in central processes stay under control of the national authority. This is a complex architectural challenge.

Global data that is collected centrally can be processed centrally as well where useful and feasible. The responsibility for process parts that apply to national statistics stays at national level. National data (like admin data / registers) needed in such processes stay under the control of the national authorities. How this would be done practically is a complex architectural challenge. Different centralisation scenarios can be chosen for different parts of the process like the negotiation phase, the collection phase and the processing phase. For global data, negotiation can best be done centrally, involving national stakeholders. Example: negotiations by ESTAT in the tourism sector.

Theme 5. Fostering cultural change

Session: 16 June 2021, 14:00-16:00 CEST

Moderator: Natalie Rosenski (Destatis, Statistics Germany)

Reporter: Sofie De Broe (CBS, Statistics Netherlands)

FACILITATORS

1) Small steps
    a. Small is beautiful: size of teams and means helps with making things happen
    b. THINK the day after tomorrow and not the hassle of the day
    c. Choose for evolution instead of revolution: step by step approach ; lower expectations
2) Bring people and ideas together
    a. Connect with colleagues and involve them
    b. Backing of innovative culture by Eurostat, the higher management
    c. Embed the existing traditional statistical activities and expertise into innovation and vice versa, e.g. integrate traditional statistical methods into the new method
    d. Bring the end users into the game of the innovation from the start
    e. Innovative and less innovative people with different skills or experience work together in a multidisciplinary team
    f. Innovation in one domain or one NSI (BD) can introduce innovation in other areas or offices
3) Grab international opportunities to collaborate and copy paste
4) Get inspired by success stories
5) See the innovation as added value/additional (so it feels less as a threat)
6) Time space to use on innovation for NSI people (e.g. 1 day a month)
7) Establish a business (start-up) approach to improve official statistics (allow being creative, making mistakes)

NEEDS

8) Provide training for new skills and hire people with new skills
9) Find subsidies (via Grants etc.) and let them be used by the department who attracted it (might require organizational changes)
10) Study the quality issues, know the limitations but use equal standards for more traditional data sources
11) Use the needs of policy makers and users as a motivator and driving force (role of the NSI)
12) Be clear on the business case : cheaper, more efficient, relevance, timeliness
13) Be aware of changes in the data culture with regards to privacy: people are willing to share with google but not with the NSI
    a. Sensitive (mobile phone) data : involve citizens, make benefit clear, educate people, address points and make transparent the purpose ; data is a public good so it should be used in this way
14) Communicate and be transparent on the purpose of using the data (targeted solutions): consider to give the data source/project a new and less threatening name

Session: 17 June 2021, 10:00-12:00 CEST

Moderator: Francis Saucy (FSO, Statistics Switzerland)

Reporter: Marc Debusschere (Statbel, Statistics Belgium)

1. Within NSIs and ESS

- **Big and pervasive problem**

Culture change is definitely a big and pervasive issue, not only relevant for the operational level of collecting data and producing statistics, but also for management, overall organisation, planning, hiring, training, …

- **Not just for the 'little' people!**

There's a tendency to only look towards the 'little' operational people to change their culture and embrace new data sources, but the problem is probably much worse and more difficult to tackle in other places in NSIs, in planning departments, methodology units, middle management and maybe even worst, in spite of ringing declarations of support for 'innovation' and 'being modern', in the highest management. Because they, mostly without being aware of it, put into place legal bases, structures, procedures and systems which seriously hinder innovation. This is the 'hard' 'frozen' culture which is often not even recognised as such, but as a given part of the environment. Both aspects, the structural character and unawareness of a problem, make this the hardest to change.

- **Frame the question correctly!**

Do not allow reluctant people to ask 'Why should we change?', put the question to them 'Why should you **not** change?'.

- **Quality is not an argument …**

Or maybe it is, but in the opposite sense! The most heard objection against using new data sources is that quality cannot be guaranteed. But this argument can easily be turned around: the current data collection and statistical production is often very far from perfect and has many quality issues. Not least of which are insufficient level of detail and timeliness, but also representativeness (some surveys have *very* low response rates …). Many new data sources offer at least the opportunity to produce much more granular and timely statistics.

2. Within society at large

- **Communication is key**

There would not be a problem if we could explain, face to face, in simple words, what we are collecting, how we do this, and most of all, why we do so. Or, in other words, if we could tell people what essential services are provided by statistics. Not just at the abstract level of 'democratic society', but also at the individual level. We should put individual aids and tools on our websites to make people aware of this (e.g., rent calculator). But even then we will still be up against bias and prejudice of individuals and groups, and against media agendas of stirring up controversy.

- **Catch 22**

We can only make the use of particular data acceptable if we can demonstrate the high value of output, but we cannot create this output as long as we are not allowed to use those data …

But maybe the virus has been helpful in forcing the issue!

- **Use cases!**

We need concrete and high-value use cases the public and decision makers want so much they do not object to accessing the data.

Theme 6. Smart statistics

Session: 16 June 2021, 14:00-16:00 CEST

Moderator: Olav ten Bosch (CBS, Statistics Netherlands)

Reporter: Kostas Giannakouris (Eurostat)

- Smart Statistics can be defined as "Multi-source multi-output statistical production systems using smart technologies, flexibly integrating data from surveys, administrative sources and various and varying new data sources"

There has been a consensus about the definition. Smart Statistics can be seen as the future of official statistics. Signs exist about continuous technological revolutions that bring changes in the official statistics world; pandemic helped in accelerating the processes in using smart technologies and developing relevant statistical products. The pandemic has brought attention to the need for more timely, more frequent and more granular data, especially relevant in times of higher uncertainty. The presence of a "killer app" that shows how smart statistics can produce very fast new indicators can help. However, there will be a long discovery path; links to official statistics still to be identified. Official statistics faced similar challenges with the use of administrative sources.

- Focus on the integration of statistics, engaging citizens, educating (the new generation of) users, and satisfying users' needs.

Statisticians should look more into what (the new generation of) users want, educate them about the potential of using smart devices, about their limitations, trigger users' needs and satisfy them. Moreover, in the context of smart statistics, statisticians should integrate the use of new data sources with the use of legacy data-collection methods. Therefore, enhance, extend, replace existing statistical products and develop new. Moreover, official statistics would be able to provide data products that would allow users (businesses) to analyse the data themselves and take informed decisions.

The participatory role of citizens should be more elaborated; give back to citizens what they expect in terms of statistical information. Official statistics have a responsibility for producing reliable statistics. Smart Statistics can be seen as an opportunity but methodological work needs to be done with respect to quality and other relevant issues. Though the fine granularity of the data would allow researchers to improve their models, it should be weighed against uncertainty and the overall quality of the final statistical product.

- "Trusted" would be an important component complementing "Smart Statistics"

"Trusted" would be important in view of the nature of the data that is re-purposed and not initially collected for statistical purposes, hence for reasons related to privacy preservation from the individual data providers' perspective. Trusted can as well be related to the role of official statistics when using new data sources related to quality, transparency, verification of statistics and the trustworthiness of the National Statistical Institutes. In that aspect, it can be a strong communication element for the institutional role of official statistics in the context of using new data sources in a new data ecosystem.

- The development of TSS strongly demands multidisciplinary skills

The development of TSS strongly demands multidisciplinary skills (statisticians, data scientists, data engineers, domain specific experts, project managers and testing capacity). Capacities to build systems are needed beyond developing tools and developing methodologies.

- Elements of a roadmap to Trusted Smart Statistics:
    - Data scouting: Identify data sources for statistical products

- Develop PoC; bring quick results and evidence of advantages
- Produce experimental statistics in collaboration with statistical production units
- Develop methodological frameworks
- Introduce changes in statistical processes (quality, integration, standards, etc.)
- Develop a vision of multi-source statistics coupled with the development of the necessary tools for flexible implementation

Various elements should be included in a roadmap to Trusted Smart Statistics. It was considered important to have a good overview of the many possible alternative new data sources for the respective statistical outputs. Bringing up advantages in terms of timelines in production of statistics would be an important element in the process. The production of experimental statistics in collaboration with statistical production units, the development of methodological frameworks and the introduction of changes in statistical processes (quality, integration) were mentioned as well. These elements should be complemented by the development of a vision of multi-source statistics with alternative scenarios (data sources) and the development of the necessary tools for flexible implementation.

- Legal provisions under European legislation for using new data sources should be part of the roadmap

Current legislation does not provide the necessary "space" to use new data sources. Legal provisions under a European legislation should be revisited along with developing the necessary technologies in particular in view of privacy preservation and data confidentiality. Along these lines, the development of policies and the extension of the code of practice should be envisaged as part of the overall process to Trusted Smart Statistics.

- Progress towards Trusted Smart Statistics is a cooperative endeavour

Collaboration in the context of the ESSnet showed the importance of a cooperative endeavour. Bringing together skills, experiences and developing together TSS should be seen as the way to go.

Session: 17 June 2021, 10:00-12:00 CEST

Moderator: Lilli Japec (SCB, Statistics Sweden)

Reporter: Barteld Braaksma (CBS, Statistics Netherlands)

Definition: 'Smart Statistics' is considered mainly a marketing term, both internal and external to the ESS. E.g. legal aspects of new data sources and combination of sources are important elements to emphasize; as well as the need to produce more timely statistics. But we have to be pragmatic! Most of all, we should keep the user perspective in mind.

Trusted: The Group was not so much in favour of adding this word. While we may have higher quality standards than some others, it is probably not effective to emphasize that in discussions. We should discuss with others outside ESS on equal terms

Domains: Where to put our effort should mainly be user-driven, we can look at the gaps in information needs and especially at new domains (green deal, digitalisation, …). It may be relevant to include also new methods like nowcasting when considering new domains of activity.

Issues: Getting legislation in place is important. We have to be patient however, and address user expectations. User expectations may differ, time series needs vs. timely data. We should move to using multiple sources (instead of silos) to become more robust. It is a fact of life that some elements are beyond our control- we need to build a stakeholder network and invest in relations management to be prepared. It helps to try to define pre-processing algorithms by ourselves to get better control on what you get. Use the strength of our collaborative mechanisms in the ESS (and beyond)!

Standardizing: Methodology and research can be jointly developed and standardised, but production should remain a national responsibility due to the national context and in particular national users' needs. International collaboration remains important; can we continue the excellent work of the ESSnets with e.g. a Center of Excellence? In addition, we wonder if a joint ESS strategy in this area has already been explicitly defined and communicated.

Other: how to implement Smart Statistics on a national level? Top level managers of the NISs need information on the advantages and disadvantages of the various organizational solutions that can be adopted within the Institutes to make room for and promote the Smart Statistics vision. Are centralized or distributed solutions better to optimize human and financial resources? Are there any pilot experiences to learn from? How can we best learn from each other?

Theme 7. Business architecture and methodological frameworks: next steps

Session: 16 June 2021, 14:00-16:00 CEST

Moderator: Tomaž Špeh (SURS, Statistics Slovenia)

Reporter: David Salgado (INE-ES, Statistics Spain)

This session followed the issues proposed in the briefing notes, namely:

- How to **promote the frameworks** which were developed?
- What is needed to **ensure continuous development** of the frameworks?
- How to identify and collect **new topics**?
- What are the **immediate next steps**?

Multiple ideas and proposals were raised which immediately were connected to each other since they may be considered from different though complementary perspectives.

Regarding the promotion of these frameworks the main points were:

1. This promotion is intricately connected to the management of cultural change. Thus, any actions on the former should be coordinated and embedded in the latter.
2. In this line of thought, the strategic promoters are identified to be the top managers of the statistical offices.
3. The dissemination and spreading of the frameworks should not be limited to 30-min presentation in a top- or intermediate-management meeting, but to provide details and hands-on experience in dedicated sessions. This dissemination should be directed both to statistical domains at the ESS level and within the Member Countries.
4. The benefits at the institutional level arising from the adoption of these frameworks should be explained and underlined.
5. Dissemination through active groups and programmes in the ESS should be undertaken: working groups on methodology and IT, ESTP programme, active support through help desk functionalities and similar,…
6. A coordinated connection and collaboration with institutions outside Europe is viewed as necessary, especially UNECE and similar organizations production statistical standards.

Regarding how to ensure continuous development, we outline:

1. The owner of these frameworks should be the ESSC, which can operates through the DIME/ITDG and other working groups and task forces.
2. In this line of thought, seeking optimal European coordination and adoption, Eurostat is thought to be playing a leading role in this development.
3. The construction of inventories, cookbooks, and toolboxes with elements from these frameworks is recognised as valuable not only for the promotion but also for their continuous development.
4. The different level of maturities between architecture and methodology should be taken into account.
5. A close contact with domain experts to identify and incorporate statistical needs into the frameworks should be stressed.
6. The ESS should provide a mechanism for the exchange of staff and for short stays in other NSIs (funding, programme,…).

Regarding new topics, many of the aforementioned points can be reconsidered under this perspective. We add:

1. We should consider a two-fold scheme going from the specific need of domain experts raised in daily production to the adaptation of new methods for the production of official statistics.
2. The communication with end-users and stakeholders should also reinforced to identify statistical needs.

For the immediate steps, we identify:

1. The ownership of the frameworks should be addressed as soon as possible.
2. Resources should be rationalised so that projects should be prioritised not to overload units and experts endangering the success of their development and implementation.
3. Connection and collaboration with other group and networks should be pursued rather than creating new ones.
4. Funding for NSIs to dedicate their experts and resources for these international projects is identified as a clear enabler.

Session: 17 June 2021, 14:00-16:00 CEST

Moderator: Anders Holmberg (ABS, Statistics Australia)

Reporter: Hannes Reuter (Eurostat)

In the roundtable discussion on the business architecture, (BA) and methodological framework (MF) a couple of topics were discussed. The topics focussed on the work already done under the ESSnet Big Data WPF and WPK and how to bring these forward. The series of topics contained:

How to promote the Frameworks, which were developed

We acknowledge the need to share information (e.g. Workshops, Videos) inside our organisations to raise awareness and a critical mass.

At the same time, we observed the need to battle proof the current frameworks further applying it in various experimental or production environments to reach higher maturity (e.g. details, practicality). The group acknowledge that only if these maturities is reached, one could propose it for “use recommendation” after an accomplished peer review.

At the same time, we identify two additional promotion ways:

a) the apparent need to create a network of leaders/users for coordination (DIME – Quality, ?? – Business architecture)
b) establish fine-grained support for implementation depending on task. This might be from the top management going all the way down to single production units or data producers.

How to ensure continuous development?

We acknowledge that the respective frameworks will only develop further if a) National Statistical Institutes and ONA’s have a keen interest and b) voluntary organisations have the resources to evaluate these in production and provide constructive feedback. The group agreed that within the ESS Eurostat needs to take on the role to be responsible for coordination and maintenance of the FW driven by the feedback from the national institutions. An update cycle should be implemented – we left the question open on which frequency.

The group question itself how the developed BA and MF can be fit into existing frameworks or structures (e.g. Centre of Excellence/DIMET/WG/TF). With this and the

additional agreement on a continuous development process, the group envisaged that resources can be saved and quality of statistics can improve.

How to identify new topics

We observe that the conventional way of exchange information (e.g. Hackathons, Tools to manage voluminous data, cutting-edge technology exchange on how we do things, conferences, Training-tour, ESTP) is one way to identify at the personal level upcoming topics. ,

At the same time, we acknowledge that a structural technological watch is missing to research/observe which topics reaching maturity. Examples mentioned were the “UN blue sky network thinking” which has working groups on e.g. synthetic data and input preserving techniques.

Immediate next steps

The group suggest as immediate steps to initiate a series of workshop Sessions – recorded for further distribution or even dedicated videos. The current practices on knowledge exchange on CROS was valued highly with the current links to various websites/information’s and especially to bibliographies for respective topics.

At NSI level these steps should be performed by voluntary countries which ‘harden’/’water proofing’ the above-mentioned BA/FA. As an outcome NSIs/ONAs should share their experience amongst their peers and document it at CROS

Finally yet importantly, the all of this will be a success if:

a) Eurostat takes on board the ownership and acts as an facilitator,
b) Discussion at DG level will be initiated

Theme 8. Further research priorities

Session: 16 June 2021, 14:00-16:00 CEST

Moderator: Marc Debusschere (Statbel, Statistics Belgium)

Reporter: Lilli Japec (SCB, Statistics Sweden)

A number of possible new data sources were identified such as Earth Observation (satellite and drones), industrial environment (farming), and services such as Airbnb, google and financials. We need a strategy in order to prioritize among these new data sources. We distinguished between supply versus demand driven projects. The strategy should be to focus on:

- External and internal demand driven pilots: user needs, knowledge gaps, to replace costly and/or burdensome surveys, improve in time and space (wider, deeper and quicker statistics)
- Supply driven pilots: “low-hanging fruits”

We discussed how to create and/or identify demand driven pilots. Most NSIs have external user groups and they could be consulted in order to identify areas where there are information gaps today. In order to create a user demand NSIs should work on marketing the data that they have and that could be combined with new data sources.

The group discussed obstacles that prevents NSIs to implement new big data sources in production and how to overcome those obstacles. A number of problems were identified and possible solutions were discussed:

- Culture. Support from top management is crucial, active involvement of subject matter colleagues is also important both on NSI and Eurostat level (e.g., subject matter working groups)
- Lack of skills on the NSI-level. Could there be a pool of top data scientists on EU-level that can support and help NSIs with implementation? Collaboration on projects within the ESS and with the ESCB, collaboration with universities (EMOS was mentioned) and other organization could be a way to handle lack of skills.
- Organization. Silos both within NSIs and on the ESS level could sometimes be a hindrance. Trying to break up these silos is important.

The following research areas for further collaboration within ESS were identified:

- Methods for integrating new and existing data sources and
- How to deal with the volatility of data. Here other sources/indicators with similar information can be used.
- Quality and robustness of data from private data sources
- Sensitivity analysis of indicators based on big data.

Division of labor within ESS was discussed. Where should data be stored and processed? We have the subsidiarity principle to adhere to.

How to disseminate the results of ESSnet Big Data I and II to a bigger audience was also discussed. In future projects it would be helpful to have one dedicated person that target the information about the results to different groups in an easily accessible way. The reports are valuable and useful but they often describe the whole process and contain detailed information on methodology. The more accessible very short summaries should “cut to the chase”.

Session: 17 June 2021, 14:00-16:00 CEST

Moderator: Alexander Kowarik (Statistics Austria)

Reporter: Dominika Nowak (Statistics Poland)

The group session was attended by 9 participants. The discussion addressed several issues related to research to enhance further exploration of new data sources and subsequently, move them from experimentation to statistical production. As a general remark the terminology issue was raised with a suggestion to clearly define the term 'smart' and/or 'new' - statistics/meters, etc.

The session led to the following conclusions:

1. Criteria to define research priorities:

A number of new data sources have been explored so far with various intensity, resulting in mixed levels of advancement and usefulness for official statistics. Hence, in order to increase the effectiveness of research efforts, application of certain criteria to define future research priorities is recommended. One of the possible criteria might involve selection of data sources with numerous potential applications, as opposed to the sources with a sole or limited application areas. Another possible approach pointed to giving priority to the data sources which allow to respond better to users' needs, which speaks to demand-driven statistics and anticipation of users' needs. Alternatively, the priority might be given to the most accessible data.

2. Cross-cutting research areas:

There are a number of cross-cutting areas to be addressed in future research. First of all, quality appeared to be a critical facet of the production of statistics with new data sources. Since quality is often raised as an argument against the employment of big data in official statistics, methodological developments in regard to quality indicators are of primary importance. Secondly, the research on architecture and standards related to new data sources were quoted an important aspect which would prepare the NSIs to react swiftly to urgent data needs (e.g. the crises), by facilitating negotiations with private data owners, as well as lowering the costs of data acquisition. Thirdly, while the difficulty with the access to certain data (e.g. MNO data, financial transactional data, etc.) is widely acknowledged, an alternative approach would involve combining and integrating data from different sources, in order to substitute statistical information produced with less accessible data. Since new data combination at the unit level is a considerable challenge, intense methodological work is required. Next to the methodology, the necessity to address implementation part, i.e. development of software and automatization of processes, designed in a modular way, in the form of micro-services was pointed out.

3. Prospective data sources and research areas:

Earth observation/satellite imagery were considered the most promising field for further research, with the highest potential to move to official production, mainly due to data accessibility. Financial transactional data were identified as the source with immense potential, however, the accessibility is the main hindrance. In addition, sensors, in terms of smart cities, smart traffic, pollution were recommended for further exploration. A demand-driven approach

for a research project was also discussed and seen important, e.g. to research new statistics supplementing the implementation of the Green Deal.

4. Organizational set-up

Formalized organization of future research on new data was underscored as a guarantee to generate progress and viable outcomes. While cooperation at various levels (national, international, regional, cross-sector, ESS, etc.) is the essential driver for development, informal collaboration, based on voluntary work was discarded and in most cases considered unproductive. Financial back-up, together with the ownership of the developments was declared critical. The inclusion of academia in future research projects is also crucial.

Plenary session: 17 June 2021, 10:00-12:00 & 14:00-16:00 CET

The plenary session was held on the third day of the event. It involved presentations of the results of all breakout group discussions, followed by an approximately 20-minute general discussion on each theme by all participants. The morning part of the session was moderated by Marc Debusschere (Deputy Project Coordinator), and the afternoon part by Peter Struijs (Project Coordinator). Final remarks were given by dr. Emanuele Baldacci (Director of methodology, IT and corporate statistical services, Eurostat), with subsequent closure by Peter Struijs. What follows is a summary of the main conclusions of the plenary discussions.

From exploration to exploitation

The role of internal and external users in developing experimental statistics was the main focus of the plenary discussion on the above-stated topic. While there was a general agreement that involvement of relevant stakeholders is crucial and it may function as a key enabler of the transformation step, the opinions differed as to whether new experimental statistics should derive from external users' needs, or be incited internally by statistics experts. It was noted that the need spurred externally facilitates the involvement and internal buy-in.

Access modalities

A common definition of experimental statistics at the European level was recognized as useful for the statistical community. In a reply to this suggestion, it was stated that experimental statistics will be addressed in currently developed innovation strategy for the ESSC. Additional remarks underscored the fact that experimental statistics should not constitute separate products, but be integrated in the official statistical production, thus, carried out jointly with production units.

Moreover, the access to a sample of data from new sources, or alternatively developing synthetic data generation models were mentioned as a potential solution to the problems with the access to some data.

The question of whether the production of statistics based on new data sources should be spurred externally or internally recurred in the plenary discussion on access modalities. It was noted that while referring to users' needs should ideally be the first step, a pragmatic approach involving both internal data mining and users' demand analysis was recommended. Another suggestion was the shift of the focus from new data sources towards innovative methodologies. It was further elaborated in a reflection on the agile development cycle, i.e. from the development of MVP (i.e. experimental statistics) through the dialog with stakeholders, constant feedback loops, and concurrent work on methodology and exploration of data sources. Such complex approach appeared the most effective, though the most challenging at the same time. Subsequent concepts raised during the discussion suggested a general shift of the focus from data towards methods. It was stipulated that low quality data may result from the fact that there have not yet been developed methods to ensure desired quality. Concurrently, it was noted that the development of the methods should be evolutionary and not a static process.

International and ESS cooperation and division of labour

The discussion concentrated on the exchange of NSIs' staff. It was pointed out that while some NSIs do not have sufficient resources for researching new data, the mobility of staff between the organizations might facilitate filling in research and competency gaps. The short-term notion of such actions should be moved from the cost generation orientation, towards their long-term perception in terms of investment. Creation of staff exchange mechanisms (similar to e.g., Erasmus for students' exchange) would be highly recommended. In that manner, not only would exchange of innovation and methods be ensured, but also peer review of developments.

Leaner and meaner statistics

Business architecture and its function as a hindrance or facilitator in the production of smart statistics span the plenary discussion summarizing the topic of leaner and meaner statistics. The importance of well-designed business architecture, adapted to the needs of actual production was emphasized. Evolutionary developments in this realm were believed to be more effective than a revolutionary approach.

Fostering culture change

The main points of the discussion addressed the role of communication and competency building. The role of demand-driven statistic was underlined, as well as the need to be cautious with placing internal focus on improving statistics above the actual user's needs. In regard to competency development and training, taking into account training needs of different age groups of statistical personnel was recommended, and the value of setting up multidisciplinary teams, not limited to data scientists, was stressed.

Smart statistics

The discussion focused on the terminology aspect of the 'trusted smart statistics' (TSS). The problem with the clarity of the term was expressed. The following logic behind its meaning was put forward: TSS should be understood as statistics that can be trusted, based on data provided by smart systems. In other words, according to the concept of the TSS, the data extracted from smart systems/smart devices is approached with official statistics' quality measures and standards.

Business architecture and methodological frameworks: next steps

It was stressed that the developments of the ESSnet BD II project in the realm of architecture, methodology and quality find interest beyond the statistical community. Hence, a dedicated action to promote and share these assets appeared relevant, also in terms of reviewing their applicability.

Future research priorities

The focus on external users' needs, already raised in the plenary discussion on other themes, was further underscored in the debate on future research priorities. It was noticed that technology and skills development should derive from the demand, not the other way round. Lack of official statistics' engagement in key European research initiatives (e.g. Destination Earth) was pointed out as a disadvantage to statistical community. In addition, searching synergies with other on-going initiatives, e.g. UNECE-ONS/DSC Machine Learning project was seen beneficial.

Final remarks by Emanuele Baldacci

Dr. Emanuele Baldacci in his final speech addressed the following issues related to the development of smart statistics and statistical innovation:

- it is important to create a joint environment for both statistical production and innovation, since the statistical product is the ultimate interest of the ESS;
- finances do not appear to be an obstacle to further research on the production of statistics with big data, as there are numerous financial frameworks at the EU level at the moment (e.g. Digital Europe program);
- collaboration, including in ESSnets, is key to the development of smart statistics in the ESS; alliances and partnerships should reach out beyond the statistical community;
- agility of official statistics is indispensable; covid pandemic was a valuable experience in this matter; the focus should now be on continuing this experience.

Closing by Peter Struijs

Peter Struijs appreciated the results of the Roundtable, which enabled the exchange of standpoints and opinions with the future outlook. The event was quoted as a spin-off of the pandemic, and since ESSnet BD II was a research project, the Roundtable can be perceived as a contribution to innovation. The value of not only the event's outputs, but more importantly, its outcomes was stressed. The project coordinator thanked the conference organisers – Marc Debusschere, Martin van Sebille, Remco Paulussen, and the Statistics Poland's team, as well as the moderators and reporters of the breakout sessions.

Annex 1. Additional issues raised in the chat

1. The discussants suggested that it would be useful to have a common definition of experimental statistics throughout the ESS which would ease communication with the stakeholders. Such definition could be implemented into the innovation strategy which is under development now.
2. Alluding to the insufficient clarity of the term *trusted smart statistics,* terminology management was stated to become essential, as it allows to speak a common language and properly understand each other. The term of innovation can be an example of misinterpretation and different understanding.
3. Experimental statistics should be more like MVPs, not as PoCs which are considered rather insufficient nowadays.
4. National Data Science Competence Centers and ESS Centers of Exellence can play an important role in facilitating the production of smart statistics. Regardless the fact how *smart statistics* are ultimately called, the official statistics should have a brand of a trustworthy source of knowledge for citizens, the economy and environment.
5. Regarding the topic of low hanging fruit, it was suggested that Airbnb could be considered in that way, especially since Eurostat started the publication process of the related data.

Annex 2. List of 8 themes discussed during the Roundtable

1. **From exploration to exploitation**
   How to implement new data sources concretely in the production of official statistics
   For what particular purpose (validation, supplementary, sole source)? What is the role of experimental statistics? What sequence of implementation steps should be followed? (e.g., exploring new data sources to gain knowledge, assess potential, learn opportunities and limits; developing adequate methodology, tools, software; formulating statistical questions based on user needs; building use cases; developing experimental statistics). Where to start? (e.g., identifying user needs, overview of new data sources, existing statistics to be made more detailed and timely)
   (Link with themes 2, Leaner and meaner statistical production, and 5, Fostering culture change)

2. **Leaner and meaner statistics**
   How to move from 'industrial' statistical production to a more flexible, rapid, just-in-time system while preserving quality and trust
   At this moment NSIs work semi-industrially, with big and complex production systems based on a fairly inflexible set of Regulations, and within rigid budgetary constraints, all of which makes it very difficult to respond in a quick and flexible way to societal developments (such as a sudden pandemic …) , additional user demands and new data opportunities. How do we achieve leaner and meaner and more opportunistic official statistics, and move away from the highly institutionalised setup now? While preserving as much as possible, however, the key elements of quality: standardisation, coordination, coherence in inputs, methods and outputs.
   (*link with themes 1, From exploration to exploitation, and 5, Fostering culture change*)

3. **Access modalities**
   How to access and use various new data sources
   Official statistics is gradually gaining experience in accessing new data sources. Various modalities can be used: pre-processing at the data source, use of 'raw' versus aggregated data; transfer of anonymised data; privacy-preserving processing; etc. What access modalities should be used under what circumstances?
   (*link with data ownership and broad access issues, not discussed in BDES 2021 because already treated elsewhere: data generated and 'owned' by private enterprises generating, data as public good versus the legitimate interests of privacy, commercial confidentiality, cost of creating, storing and querying data, ...*)

4. **International and ESS cooperation and division of labour**
   Subsidiarity: collaboration and division of labour on new data sources between global, ESS, national, regional and city levels.
   How to divide the considerable amount of the work on new data sources between international statistical organisations (Eurostat, UNSTAT, UNECE, OECD) and NSIs or agencies, EU and non-EU (ESS NSIs, EC Directorates, US agencies, StatCan, StatAU, ONS, …)? Do we specialise, and if so, how?
   How to organise the work within the ESS, at the exploratory, pilot and statistical production stages, for the data collection and processing? Should specific data types such as AIS, sharing platform data, data from the big tech companies or multinational network operators be collected at the European level and what systems should be set up for them? Orr how, alternatively, can collection be coordinated for data relevant for several NSIs?
   (*link with participation in international groups and projects and the architecture of ESS collaboration, the creation of an ESS Trusted Smart Statistics (TSS) centre and hubs, and the more general subsidiarity between Eurostat, NSIs, various other agencies, whether ONA or not (national banks, domain specialists), regional statistical offices and smart city initiatives*)

5. **Fostering culture change**
   How to accomplish the considerable culture change needed, within statistical institutes and in society at large, to use new data sources for official statistics.
   Within statistical institutes the switch to 'non-traditional' data sources requires that managers and domain statisticians leave the comfort zones of traditional sources and long-tested methods and procedures and to move in uncharted territories of new methods. How can this be achieved? What are the facilitators and obstacles? How are we doing on this? At the same time a culture change outside of statistics is needed. How do we convince society of the importance of new data sources for policy, citizens, enterprises, research and of the necessary balance to be struck with other concerns such as privacy, commercial confidentiality, data ownership and 'exclusivity' rights? How do we communicate with policy makers, the media, different user categories to get this across? How do we convince data owners they are required to contribute to the public good?
   (*Link with theme 1, From exploration to exploitation, and 2, Leaner and meaner statistics*)

6. **Smart statistics**
   Multi-source multi-output statistical production systems, flexibly integrating data from surveys, administrative sources and various and varying new data sources
   Is there agreement on this preliminary definition? And what does it mean concretely, operationally and exactly? How will smart statistics look like in statistical production in various domains, and how will we get there in practice? How will the problems of stability and continuity inherent in many new data sources be handled? How to solve the methodological issues of integration and stable output with varying/unstable inputs? How can smart statistics be standardised across the ESS and over time? How to relate to other involved parties and intermediaries (e.g., . the providers of smart farming systems often receiving sensor data and essential for standardising)?
   (*link with themes 1, From exploration to exploitation, and 2, Leaner and meaner statistics*)

7. **Business architecture and methodological frameworks: next steps**
   How to follow up on the work done by ESSNet Big Data WPF and WPK?
   State of affairs and next steps for the frameworks developed by WPF and WPK, their role and integration in the ESS, and future work in this respect (e.g., integrating to other data sources.

8. **Further research priorities**
   What further research is needed to move various new data sources to experimental statistics and statistical production?
   First of all, what new data sources not yet under investigation should be considered for (first) exploration and lead to future new pilots? Which data sources already considered most merit further research to move them from the exploratory and pilot to the implementation stage? The ESSnet Big Data II Pilots Track workpackages? (viz financial transaction, earth observation, mobile network data and innovative tourism statistics) On what aspect(s), for each data source or workpackage, should be the focus? (e.g., methodology, pilots, techniques, standards, …) What is the order of priority of sources and issues, in view of the prospects of moving to implementation?
   (*link with theme 1, From exploration to exploitation*)